| INCH-POUND |

MIL-E-17814F
1 June 1992
SUPERSEDING
MIL-E-17814E
21 March 1984

MILITARY SPECIFICATION

EXPANSION JOINTS, PIPE, SLIP-TYPE, PACKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers guided packed slip-type expansion joints used to absorb only axial movement in piping systems. This movement may be caused by thermal expansion, contraction or other factors.

1.2 Classification. The expansion joints will be of the following type and classes as specified (see 6.2).

Type IV - Internally-externally guided; injected semiplastic type packing.

Class 1 - Single slip-tube.
Class 2 - Double slip-tube.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-18T10:38Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (except for Systems Indicated Herein); Packaging of.
MIL-F-3541 - Fittings, Lubrication.

(Unless otherwise indicated, copies of military specifications are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Steel Pipe Flanges and Flanged Fittings.
ASME B16.9 - Factory-Made Wrought Steel Buttwelding Fittings.
ASME B31.1 - Power Piping.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

ASTM

ASTM A 53 - Pipe, Steel, Black and Hot Dipped Zinc-Coated Welded and Seamless.
ASTM B 650 - Electrodeposited Engineering Chromium Coatings on Ferrous Substrates.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard commercial product. The expansion joint shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of

Source: https://assist.dla.mil -- Downloaded: 202610-18T10:38Z
Check the source to verify that this is the current version before use.

the manufacturer's standard commercial product, shall be included in the expansion joint being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures, representing the latest production model.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.3.1 Joint materials. Expansion joints shall be constructed of one or more of the following materials: Steel, ductile iron, or other alloys as specified (see 3.6, 6.2 and 6.6).

3.4 Description. The expansion joint shall consist of a single or double slip-tube arranged to slide within the body of the joint. As specified herein, the expansion joint shall be provided with flanged or welded end connections, slip-tube guides, limit stops, tie-rods, stuffing boxes, packing glands, base, lubrication connections and fittings, service outlet, drain connectors, vent connection, and the required bolts, studs, and nuts. The expansion joint shall be the size specified.

3.4.1 Type IV. Joints shall have slip-tube guides of one piece formed or welded construction integral with the joint stuffing box. The injectable packing shall be contained in the joint stuffing box by packing rings at each side of the injectable packing zone in accordance with 3.6.13. The primary packing for this type of joint shall be of the pressure injected type suitable for injection under pressure through packing injection devices located radially about the joint stuffing box in accordance with 3.6.14. Limit stops shall be furnished in accordance with 3.6.5.

3.4.2 Classes.

3.4.2.1 Class 1. Class 1 expansion joints shall have a slip-tube at only one end of the joint.

3.4.2.2 Class 2. Class 2 expansion joints shall have slip-tubes at both ends of the joint, with mandatory intermediate anchor base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

3.5 Performance. The joint shall be capable of absorbing the total axial expansion movement specified (see 4.5.2 and 6.2), and shall be designed to require the minimum of service maintenance when performing the function for which it is intended.

3.5.1 Service conditions. The expansion joint shall be suitable for operation under the following types of service, as specified (see 6.2):

a. Temperature-range to include installation, minimum, and maximum temperature.
b. Pressure-operating pressure and maximum shock pressure.
c. Type of material to be conveyed.
d. Velocity of material being conveyed.
e. Corrosive conditions.
f. Special conditions peculiar to installation.

3.5.2 Compression force. Unless otherwise specified (see 6.2), the force required to compress or extend the expansion joint shall not be greater than 1,250 pounds per inch of pipe diameter.

3.5.3 Hydrostatic pressure. Expansion joints up to and including 24 inches nominal pipe size shall be capable of withstanding, without damage, a hydrostatic pressure equal to 1-1/2 times the rated pressure of the joint. Joints larger than 24 inches shall be tested at a pressure to be specified by the procuring activity (see 6.2).

3.6 Construction. The expansion joints covered by this specification shall be designed and constructed in accordance with the practice set forth in ASME B31.1, where applicable.

3.6.1 Body. Unless otherwise specified (see 6.2), the body of the joint shall be made of ductile iron, cast steel, or fabricated steel, conforming to applicable ASTM standards for the pressure, temperature, and other service conditions. When the body is made of ductile iron, only the slip-tube end may be beveled for welding.

3.6.2 Gland. Unless otherwise specified (see 6.2), the gland of the joint shall be made of ductile cast iron, cast steel, or fabricated steel, conforming to applicable ASTM standards for the pressure, temperature, and loading of this member.

3.6.3 Slip tube. The slip-tube shall be manufactured of steel complying with ASTM A 53, type S, grade A or B. As a minimum (see 6.2), it shall be manufactured of Schedule 80 for all sizes to 16 inches inclusive and Schedule 60 for sizes 18 to 24 inches.

3.6.3.1 Chromium-plating. The sliding slip-tube shall be plated with a minimum of 2 mils of engineering chromium per ASTM B 650.

3.6.4 End connections. The end connections of the joint shall be provided with pipe flanges or welding nipples, as specified (see 6.2). Flanges shall conform to ASME B16.5 and shall be of the size, material, pressure class, and

Source: https://assist.dla.mil -- Downloaded: 202640-18T10:38Z
Check the source to verify that this is the currentversion before use.

facing specified. Welding nipples shall be beveled for welding to conform to ASME B16.9 and shall be of the size, material, and nominal wall-thickness designation specified.

3.6.5 Limit stops. Limit stops may be of either the external or internal type, and shall be designed to withstand the full line pressure in all normal conditions of loading encountered during joint operation and testing, including line pressure load resulting from failure of pipeline anchors. Tie-rods may be used as external limit stops.

3.6.6 Guides. Slip-tube guides shall be designed to effectively maintain the slip-tube concentric with the stuffing box throughout the specified traverse.

3.6.7 Stuffing box. The stuffing box shall be sufficiently deep and proportioned to hold ample packing in accordance with paragraph 3.6.14 to assure a tight seal with minimum replacement or addition of packing. The design of the stuffing box shall be such that ring packings will not readily be extruded from the stuffing box and injectable packing used for sealing will not readily be extruded.

3.6.8 Base. When specified (see 6.2), class 1 expansion joints shall be provided with a base. The base shall be designed as a main anchor. All expected forces and moments for the bases must be specified. Class 2 expansion joints shall be provided with a base suitable as an intermediate anchor. The base shall be made of ductile iron, cast steel, or fabricated steel conforming to the applicable ASTM standards and shall be suitably attached to the joint, and drilled for the specified bolting.

3.6.9 Lubrication fittings. When specified, lubrication fittings shall be provided. Lubrication fittings shall conform to MIL-F-3541 or shall be the button-head type. When the joint is to be insulated, lubrication fittings shall be provided with extension nipples of suitable length for the thickness of the insulation specified (see 6.2).

3.6.10 Service outlet. When specified (see 6.2), a service outlet shall be achieved by the use of weldolet, threadolet, or saddleolets as specified. The service outlet shall permit unobstructed access, regardless of the position of slip-tubes.

3.6.11 Drain connections. When specified (see 6.2), each joint body shall be provided with a tapped and plugged drain connection. The drain connection shall be located between the rear of the slip-tube piston and the bottom of the stuffing box.

3.6.12 Bolting. Bolting required as an integral part of the joint shall be provided. Bolts, half studs, and accompanying nuts required as an integral part of the joint shall conform to the requirements of 3.6.

3.6.13 Packing. The packing shall be provided and installed by the manufacturer. All packing materials shall be suitable for the specified service and shall not react chemically with the components of the joint or the media being conveyed. Injectable packing shall be self-lubricating flake graphite suitable to 850 degrees Fahrenheit (see 6.2). Unless otherwise specified,

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

joints shall be furnished with spare injectable packing in plug form for each packing device at each joint.

3.6.14 Packing injection device. To permit packing injection under full line pressure the expansion joints shall be equipped with packing injection device(s) welded to the joints' stuffing box and designed with heavy duty acme threads and furnished complete with a mating plunger. To prevent blow back of the injectable packing and/or flowing fluid, the discharge tip of the packing injection device shall be designed to produce a check-valve effect when plunger is removed (see 6.2).

3.7 Treatment and painting. Except for flanged faces and threaded connections, all external surfaces subject to corrosion shall be treated and painted in accordance with the manufacturer's standard practice.

3.8 Repair or maintenance and accessories. Repair or maintenance parts and accessories as specified (see 6.2), shall be furnished.

3.9 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers required as an integral part of the joint shall be provided in accordance with good commercial practice and all bolts and nuts shall be completely fastened and tight.

3.10 Workmanship. Workmanship shall be in accordance with the best commercial practice for this type of product. All expansion joints shall be clean and free of burrs, cracks, pits, porosity, and other imperfections affecting the serviceability of the product.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the expansion joint when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5, and packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Every expansion joint shall be examined as specified in 4.4. Sampling for tests specified in 4.5 and packaging inspection specified in 4.6 shall be as agreed upon by contracting agency and contractor.

4.4 Examination. Each expansion joint shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. The first article shall be tested as specified in 4.2.1. Each production unit shall be tested as specified in 4.5.1 through 4.5.3. Failure to pass any test shall constitute cause for rejection.

4.5.1 Hydrostatic test. Each test joint sample up to 24 inches in nominal pipe size shall be subjected to a hydrostatic pressure equal to 1-1/2 times the rated pressure of the joint. The pressure shall be maintained for 60 seconds. This test shall be performed with the joint extended. Joints larger than 24 inches nominal pipe size shall be subjected to pressure tests specified by the agency concerned (see 3.5.3).

4.5.2 Axial movement. Each test sample shall be tested to determine that the unit meets the specified axial-movement and axial-force requirement (see 3.5, 3.5.2, and 6.2).

4.5.3 Materials. Materials shall be tested in accordance with the applicable ASTM standard (see 3.6.3 and 6.6).

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-V-3 with the level of preservation and the level of packing as specified (see 6.2).

5.2 Preparation for shipment. Expansion joints shall be prepared for shipment in a manner that maintains the proper end-to-end dimensions required for installation of the joint. When no installation temperature is specified, the expansion joint shall be prepared for shipment with end-to-end dimensions in accordance with the manufacturer's standard practice. Each joint shall be accompanied by a complete set of instructions for compression or extension to the correct end-to-end dimension of the joint at the existing temperature during installation. These instructions shall be attached to the joint and shall be fully protected against moisture.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Expansion joints covered by this specification are intended for steam, water, oil, or gas piping services which require the absorption of movement resulting from the thermal expansion of piping or other factors.

6.1.1 Service conditions. Slip-tube expansion joints are suitable for pressure-temperature service conditions in accordance with the ratings prescribed by the ASME piping standards for the type of materials used.

6.1.2 Types. No expansion joint listed in this specification is intended for operation without adequate provision for guiding and anchoring the system in accordance with the manufacturer's recommendations and in accordance with standard piping practices (see 6.2).

6.1.3 Sizes. Expansion joints most often are custom engineered and fabricated for specific applications. Because there is an infinite number of sizes that might be required for use in piping systems which include both standard and nonstandard sizes, specific sizes are not cited in the classification paragraph of this specification (see 1.2 and 6.2).

6.1.4 Guides. Packed joints have slip-tube guides provided both on the internal (pressure) and external (atmospheric) side of the packing chamber. The external guide is either integral with or rigidly attached to the body. The packing for this joint is the semiplastic type which can be added while the joint is under full line pressure. To completely repack the joint, however, the service must be shut down.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class, and size required (see 1.2, 6.1.2, and 6.1.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. First article if required and approval (see 3.2, 4.2.1, and 6.5).
e. Joint materials other than as specified (see 3.3.1 and 6.6).
f. Total axial movement (see 3.5).
g. Service conditions (see 3.5.1 and 4.5.2).

(1) Temperature (degrees Fahrenheit): maximum and minimum operating temperature and superheat.
(2) Pressure (pound-force per square inch), whether constant, pulsating, or shock.
(3) Fluid conveyed.
(4) Internal or external corrosive conditions if applicable.

h. Compression force, if other than specified (see 3.5.2).
i. Test pressure for joints larger than 24-inch pipe size (see 3.5.3).
j. Whether the body, gland, and sleeve shall be constructed other than as specified (see 3.6.1, 3.6.2 and 3.6.3).
k. Type end connection: for flanged end connections specify pressure class, facing, and drilling required. For welding ends specify pipe schedule (or wall thickness) of nipples (see 3.6.4).
l. If anchor base is required for class I expansion joints (see 3.6.8).
m. The expected forces and movements (see 3.6.8) if anchor base is required for class I expansion joints. Lubrication fitting if required (see 3.6.9).
n. Service connection orientation of outlet connection and type of connection if required (see 3.6.10).
o. Service outlet arrangement of outlet connection and type of connection if required (see 3.6.10).
p. If a drain connection is required (see 3.6.11).
q. Bolting and other hardware required (see 3.6.12).
r. Packing, as specified (see 3.6.13).
s. Packing injection device check valve, if required (see 3.6.14).
t. Repair or maintenance parts, tools, and accessories required (see 3.8).
u. Level of cleaning, drying, preservation, and packaging and level of packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 Drawings. A dimensional sketch or drawing in sufficient detail to describe the components the manufacturer proposes to furnish should be submitted to the agency concerned with the bid proposal. These sketches or drawings should be accompanied by a copy of the manufacturer's specification and rating data providing sufficient detail to signify compliance with the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-18T10:38Z
Check the source to verify that this is the current version before use.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one expansion joint. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.6 Suggested construction materials. Materials from which expansion joints are constructed may include, but are not limited to, the materials called out in the following:

ASTM A 36 - Structural Steel.
ASTM A 53 - Pipe, Steel, Black and Hot Dipped Zinc-Coated, Welded and Seamless.
ASTM A 106 - Seamless Carbon Steel Pipe for High Temperature Service.
ASTM A 285 - Pressure Vessel Plates, Carbon Steel, Low and Intermediate Tensile Strength.

6.7 Supersession data. Users are directed to Military Specification MIL-E-17813 on bellows expansion joints for the internally - externally guided permanent packless (bellows) seal joint, formerly referred to as type V in previous editions of this specification. Types I, II and III have been eliminated from this specification.

6.8 Subject term (key word) listing.

Anchor gland
Base guides
Fittings force, compression
Glands, packing
IFE
Joint, expansion
Packing
Plating, chromium
Pressure, hydrostatic
Slip, single
Slip, double
Stops, limit
Tube slip

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to extensive change.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review Activities:
Air Force - 82
DLA - CS

User Activity:
Army - CE

Preparing Activity:
Navy - YD

(Project 4730-1130)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-E-17814F	920601

3. DOCUMENT TITLE

EXPANSION JOINTS, PIPE, SLIP-TYPE, PACKED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
G. M. KRALIK	805-982-5741	551-5741
c. ADDRESS (Include Zip Code) COMMANDING OFFICER NAVAL CONSTRUCTION BATTALION CENTER (1564R) PORT HUENEME, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil - Downloaded: 2026-10-18T10:38Z
Check the source to verify that this is the current version before use.